Metallic Bonding

Exam code: 0620 & 0971

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Updated on 30 September 2025

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Metallic bonding

The structure of a metal

Extended tier only

- Metals consist of **giant** structures
- Within the metal lattice, the atoms lose their outer electrons and become positively charged metal **ions**
  - The outer electrons no longer belong to any specific metal atom and are said to be **delocalised**
  - This means they can move freely between the positive metal ions and act like a *"sea of electrons"*
- The metallic bond is the strong force of **attraction** between the positive metal ions and the delocalised electrons
- This type of bonding occurs in metals and metal alloys, which are mixtures of metal

Diagram to show metallic bonding

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

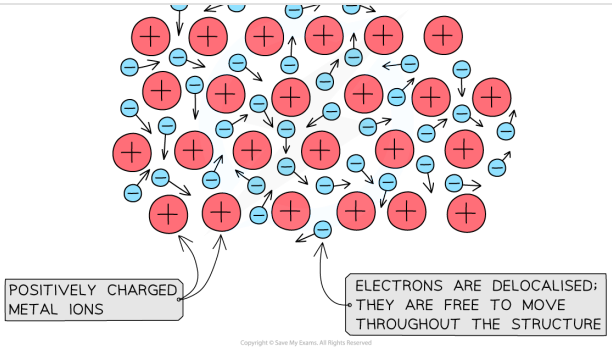


Diagram showing metallic lattice structure with delocalised electrons

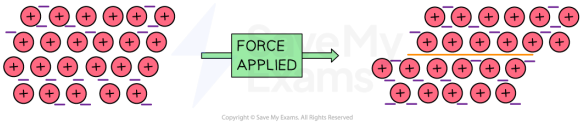
Properties of metals

What are the properties of metals?

Extended tier only

- Most metals have **high melting and boiling points**
  - There are strong electrostatic forces of attraction between the positive metal ions and the negative delocalised electrons within the metal lattice structure
  - These needs lots of energy to be broken
- Metals are **good conductors of heat and electricity**
  - The delocalised electrons are free to move and carry a charge through the whole structure
- Most metals are **malleable**
  - This means they can be hammered into shape
  - This is because the atoms are arranged in layers which can slide over each when force is applied

Malleability of metals



When a force is applied, the layers of positive ions slide over each other

- Most metals are **ductile**
  - This means they can be pulled into wires
  - This is also because the atoms are arranged in layers which can slide over each when force is applied

careful of the terminology you use. Don't get confused with ionic compounds.

Metals can conduct electricity as they have free **electrons** that can carry charge whereas molten or aqueous **ionic** compounds can conduct electricity because they have free **ions** that can carry charge.

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